**Term Project Deliverable #2: Microservice Design with Collaborations**

**Project:** CampusKit -- peer-to-peer short-term equipment rental for university students

**1. Core Business Use Cases & Actors**

- **UC-1: Publish an Equipment Listing** (Primary Actor: Owner)
- **UC-2: Book and Pay for a Rental** (Primary Actor: Renter)
- **UC-3: Complete the Handover, Return, and Settlement** (Primary Actors: Owner and Renter)

**2. External Systems**

- **Payment Gateway (Omise / Stripe):** Reached via PaymentGatewayAdapter.
- **University Identity Provider:** Reached via UniversityIdpAdapter.
- **Email & Push Providers:** Reached via adapters in the Notification Service.
- **Object Storage (S3):** Clients interact directly via short-lived signed URLs minted by Services (zero bytes flow through our microservices).

**3. Service, Operations, and Collaborators**

This table maps every system operation to the service that owns its business capability, meeting the requirement that logic resides in the proper domain. The API Gateway routes requests but contains no business logic. *(Scheduled)* indicates a time-triggered background job.

| Service & Responsibility | Key Operations | Collaborators (Calls Made) |
|---|---|---|
| **Student Identity Service**<br>Owns verification, account standing, and eligibility.<br>Private DB: identity_db | verifyStudentIdentity()<br>checkRentalEligibility()<br>applyAccountRestriction()<br>requestErasure()<br>purgeExpiredIdentityDocuments() *(scheduled)* | **Rental / Payment / Review:** pseudonymiseParty()<br>**Notification:** sendNotification()<br>**External:** IdP, Object Storage |

| Service & Responsibility | Key Operations | Collaborators (Calls Made) |
|---|---|---|
| **Listing Service**<br>Owns items, prices, condition evidence specs, and screening. Private DB: listing_db | createListing()<br>searchListings()<br>quoteBooking()<br>moderateListing() | **Student Identity:** checkRentalEligibility()<br>**Payment:** getPayoutDestinationStatus()<br>**Notification:** sendNotification()<br>**External:** Object Storage |
| **Booking Service**<br>Owns availability calendar and commercial agreements. Private DB: booking_db | findAvailableListings()<br>requestBooking()<br>approveBooking()<br>cancelBooking()<br>expireStaleRequests() *(scheduled)*<br>reconcileCalendar() *(scheduled)* | **Listing:** getListingDetail(), quoteBooking()<br>**Payment:** authorizeRentalPayment(), voidAuthorization()<br>**Rental:** createRental(), cancelRental()<br>**Identity / Notification** |
| **Rental Service**<br>Owns custody, handovers, overdue states, and settlement orchestration. Private DB: rental_db | createRental()<br>issueHandoverCode()<br>confirmPickup()<br>confirmReturn()<br>submitConditionReport()<br>holdSettlement()<br>closeRental() *(scheduled)* | **Payment:** captureRentalFee(), settleDeposit(), payoutToOwner()<br>**Review:** openReviewWindow()<br>**Notification / Object Storage** |
| **Payment Service**<br>Owns money movement and the internal ledger. Private DB: payment_db | savePaymentMethod()<br>authorizeRentalPayment()<br>captureRentalFee()<br>settleDeposit()<br>payoutToOwner()<br>reconcileLedger() *(scheduled)* | **External:** Payment Gateway (Authorize, Capture, Void, Transfer)<br>**Notification** |

| Service & Responsibility | Key Operations | Collaborators (Calls Made) |
|---|---|---|
| **Review and Reputation Service**<br>Owns two-sided reviews. Private DB: review_db | submitReview()<br>getReputationSummary()<br>recordDisputeOutcome() | **Notification:**<br>sendNotification() |
| **Dispute Resolution Service**<br>Owns damage claims and adjudication. Private DB: dispute_db | openDispute()<br>submitEvidence()<br>resolveDispute()<br>enforceDisputeCompliance() *(scheduled)* | **Rental:** getRental(), holdSettlement(), closeRental()<br>**Identity:** applyAccountRestriction()<br>**Payment / Review / Listing / Notification** |
| **Notification Service**<br>Owns message delivery. Private DB: notification_db | sendNotification()<br>upsertNotificationContact()<br>retryFailedDeliveries() *(scheduled)* | **External:** Email Provider, Push Provider |

## 4. Architecture Diagram (C4 Level 2: Containers)

